

# FARJANA SULTANA SAMIA

AI/NLP Researcher — Ph.D. Student, Iowa State University

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## PROFESSIONAL SUMMARY

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Ph.D. researcher specializing in NLP, Large Language Models, and Retrieval-Augmented Generation (RAG) with a year of industry experience deploying NLP systems at production scale. Passionate about applying foundation models and LLM techniques to complex, real-world systems.

## EDUCATION

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**Ph.D. in Computer Science** — Iowa State University, Ames, IA

*Jan 2024 – Present*

- GPA: 3.92 / 4.00

**B.Sc. in Computer Science and Engineering** —

*Feb 2017 – Dec 2021*

Manarat International University, Bangladesh

- GPA: 3.90 / 4.00 — Graduated with Distinction; 1<sup>st</sup> rank among 48 graduates

## RESEARCH EXPERIENCE

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**Graduate Researcher Assistant** — **Data and Knowledge Mining Lab** —

*Jul 2024 – Present*

Iowa State University, Ames, IA

- Conducting advanced research on NLP and LLMs including transformer/foundation models, knowledge graphs, and RAG pipelines for complex information retrieval tasks
- Building Entity Linking (EL) and Relation Extraction (RE) systems that integrate structured knowledge with language models
- Applying prompt engineering and fine-tuning of foundation models to solve domain-specific language understanding problems

## PUBLICATION

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**Bridging Lexical Divergence: LLM-Assisted, Cost-Efficient, Zero-Shot Scientific Entity Linking** *Findings of EMNLP 2026*

**Contribution:** Developed an annotation-free pseudo-labeling and filtering pipeline for model fine-tuning.

## PROFESSIONAL EXPERIENCE

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**Graduate Teaching Assistant** — (COM S 327, COM S 227) —

*Jan 2024 – Present*

Iowa State University, Ames, IA

- Advanced Programming Techniques: mentoring students in algorithm design, debugging, memory management, pointers, templates, and STL (C++)
- Object Oriented Programming(OOP): led lab sessions, supported students with Java and OOP concepts, graded assignments, and collaborated with instructors

**Software Engineer (NLP)** — Technometrics Limited, Dhaka, Bangladesh

*Dec 2022 – Dec 2023*

- Built Bengali multi-label topic classification model using fine-tuned BERT (Flair, PyTorch, FastText) achieving 85% accuracy across 17 topic classes

- Developed Bengali aspect-based sentiment analysis system (PyABSA) to predict per-aspect polarity scores within a single sentence
- Engineered social media and news scraping bot (Selenium, MongoDB, Multiprocessing) with automated scheduling and REST API delivery; includes Word Cloud, trending topics, and viral post analytics
- Built scalable scraping pipeline covering 25 Bengali blogs and 15 news portals using Scrapy and MongoDB

## KEY PROJECTS

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### Advanced RAG System with Multi-Query & Contextual Compression

- Designed RAG pipeline using GPT and LLaMA with chunking, re-ranking, and contextual compression to improve LLM response accuracy
- Deployed as an interactive web app using Funix, enabling non-technical stakeholders to query the system via browser

### Sentiment Analysis with Aggregated Token Attention Network (ATAN)

- Designed multi-model sentiment classifier on Amazon Fine Food Reviews using BERT variants (BERT-Base, DistilBERT, multilingual); achieved 77% accuracy via token-level attention aggregation
- Applied lemmatization, class balancing (60K samples, 3 classes), and feature engineering for optimized performance

### Object Recognition – CIFAR-10 & Red Blood Cell Counting via Transfer Learning

- CNN-based image classification on CIFAR-10 (60K images, 10 classes) using Keras; cell detection using Faster RCNN, SSD, and YOLO — demonstrates versatility beyond NLP

## TECHNICAL SKILLS

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**Languages:** Python (expert), C/C++, C#

**AI/ML:** PyTorch, TensorFlow, Keras, HuggingFace Transformers, LangChain, LlamaIndex, FAISS

**NLP/LLM:** BERT, DistilBERT, GPT, LLaMA, RAG, Entity Linking, Knowledge Graphs, Prompt Engineering, Model Fine-Tuning

**Web & APIs:** Flask, Django, FastAPI, HTML/CSS, JavaScript, Elasticsearch, SQL

**Databases:** MySQL, MongoDB, Neo4j (Graph DB), Pinecone (Vector DB), Semantic Search

**Scraping:** Scrapy, Selenium, BeautifulSoup, Requests

**Tools & OS:** Git, Jupyter, VS Code, PyCharm, Linux (Ubuntu)